Aim:

To configure and implement Virtual Local Area Networks (VLANs) on a Cisco switch and verify communication between devices belonging to the same VLAN.

Description:

VLAN (Virtual Local Area Network) is a logical division of a physical switch into multiple independent networks. In this experiment, a Cisco 2960 switch is configured with two VLANs:

- **VLAN 10 – CSE**
- **VLAN 20 – EEE**

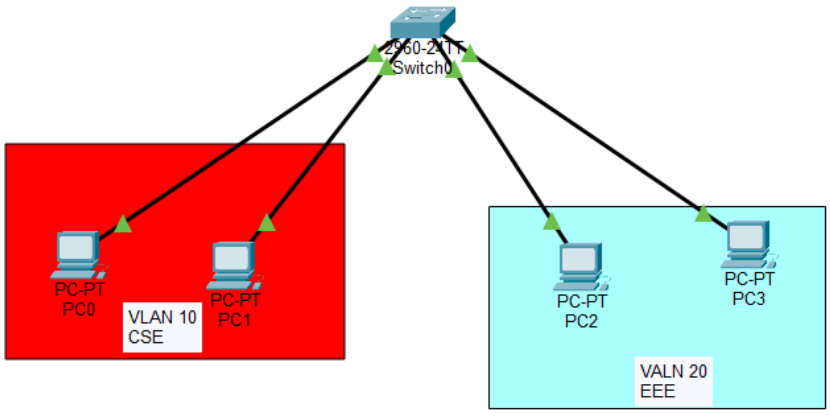
Four PCs are connected to the switch. PC0 and PC1 are assigned to VLAN 10, while PC2 and PC3 are assigned to VLAN 20. Devices belonging to the same VLAN can communicate with each other, whereas devices belonging to different VLANs cannot communicate directly without inter-VLAN routing.

Procedure:

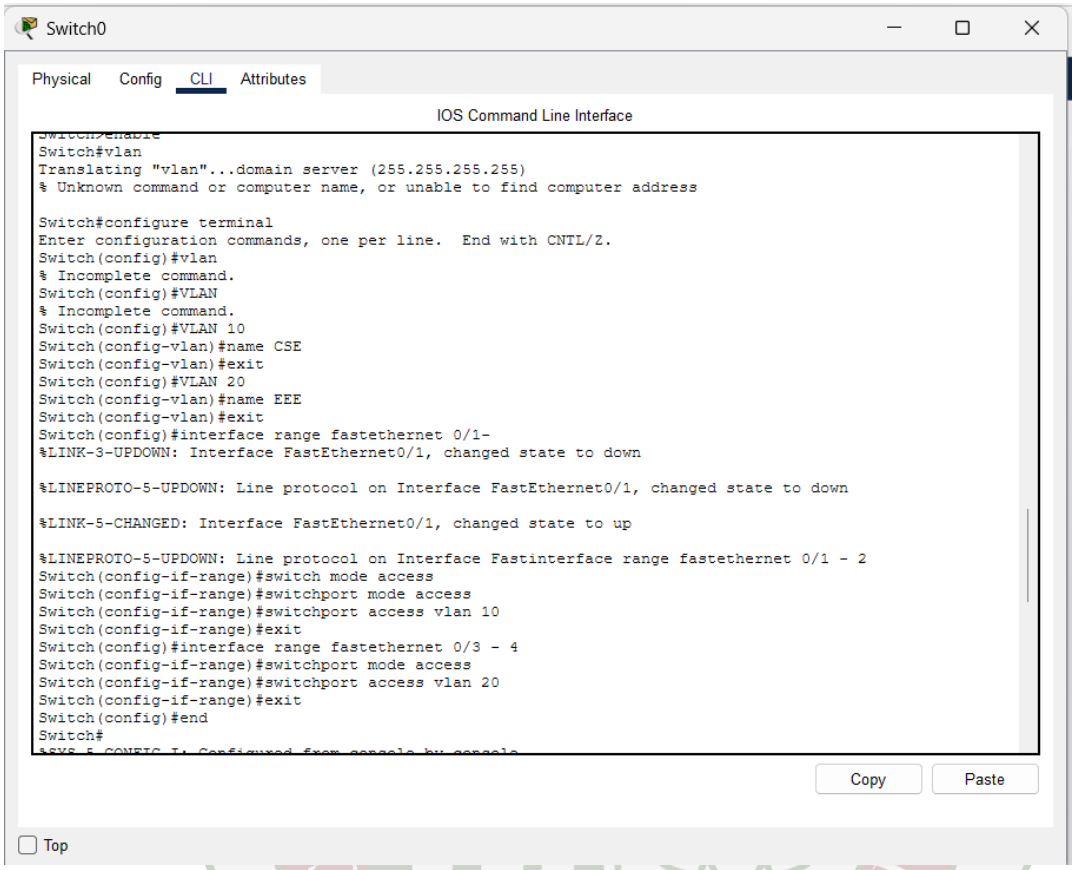
1. **Open Cisco Packet Tracer.**
2. Place **1 Cisco 2960 switch** and **4 PCs** in the workspace.
3. Connect the PCs to the switch using **Copper Straight-Through cables**.
4. Assign the ports:
    - PC0 → Fa0/1
    - PC1 → Fa0/2
    - PC2 → Fa0/3
    - PC3 → Fa0/4
5. Assign IP addresses:
    - PC0 → 192.168.10.1
    - PC1 → 192.168.10.2
    - PC2 → 192.168.20.1
    - PC3 → 192.168.20.2
    - Subnet mask → 255.255.255.0
6. Open the **Switch CLI** and enter privileged mode using:
   enable

7. Enter configuration mode:
   configure terminal
8. Create **VLAN 10** and name it **CSE**.
9. Create **VLAN 20** and name it **EEE**.
10. Assign **Fa0/1 and Fa0/2** to VLAN 10.
11. Assign **Fa0/3 and Fa0/4** to VLAN 20.
12. Save the configuration using:
    write memory
13. Verify the VLAN configuration using:
    show vlan brief
14. Test communication between **PC0 and PC1**.
    → Communication should be **successful** because both belong to VLAN 10.
15. Test communication between **PC2 and PC3**.
    → Communication should be **successful** because both belong to VLAN 20.
16. Test communication between **PC0 and PC3**.
    → Communication should **fail** because they belong to different VLANs.

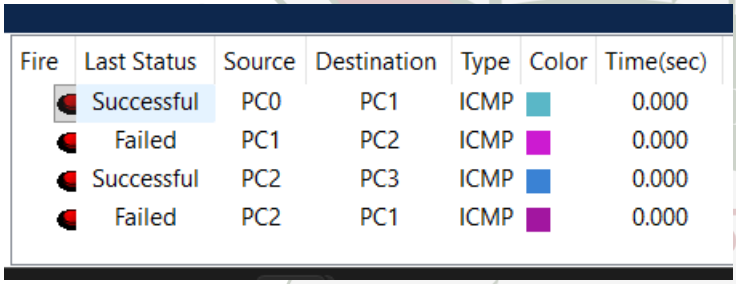
NETWORK TOPOLOGY



SWITCH 0 : commands for create vlan and access switch



Output:



Fire	Last Status	Source	Destination	Type	Color	Time(sec)
	Successful	PC0	PC1	ICMP		0.000
	Failed	PC1	PC2	ICMP		0.000
	Successful	PC2	PC3	ICMP		0.000
	Failed	PC2	PC1	ICMP		0.000

Justification:

VLANs are used to divide a network into separate logical groups without requiring separate physical switches. This improves network security, reduces broadcast traffic, simplifies network management, and provides better isolation between different departments or groups.

In this experiment, VLAN 10 and VLAN 20 demonstrate how users can be separated logically on the same physical switch. Communication within the same VLAN is successful, while communication between different VLANs is blocked unless a router or Layer-3 switch is configured.

Aim:

To configure and implement VLANs on two Cisco switches and establish communication between devices belonging to the same VLAN across the switches.

Description:

VLAN (Virtual Local Area Network) is a logical division of a physical network into separate broadcast domains. In this experiment, two Cisco 2960 switches are connected using a trunk link.

Procedure:

1. Open Cisco Packet Tracer.
2. Place 2 Cisco 2960 switches and 4 PCs in the workspace.
3. Connect:
    - PC0 → SW1 Fa0/1
    - PC1 → SW1 Fa0/2
    - PC2 → SW2 Fa0/1
    - PC3 → SW2 Fa0/2
4. Connect SW1 Fa0/24 to SW2 Fa0/24 using a copper cable. This will be the trunk link.
5. Assign IP addresses:
    - PC0 → 192.168.10.1
    - PC1 → 192.168.10.2
    - PC2 → 192.168.20.1
    - PC3 → 192.168.20.2
    - Subnet mask → 255.255.255.0
6. Open the CLI of SW1 and enter configuration mode.
7. Create VLAN 10 (STUDENTS) and VLAN 20 (FACULTY).
8. Assign Fa0/1 to VLAN 10 and Fa0/2 to VLAN 20 on SW1.
9. Configure Fa0/24 of SW1 as a trunk port.
10. Open the CLI of SW2.
11. Create the same VLAN 10 (STUDENTS) and VLAN 20 (FACULTY) on SW2.
12. Assign Fa0/1 to VLAN 10 and Fa0/2 to VLAN 20 on SW2.

13. Configure Fa0/24 of SW2 as a trunk port.

14. Save the configurations on both switches using:

write memory

15. Verify VLANs on both switches using:

show vlan brief

16. Verify the trunk connection using:

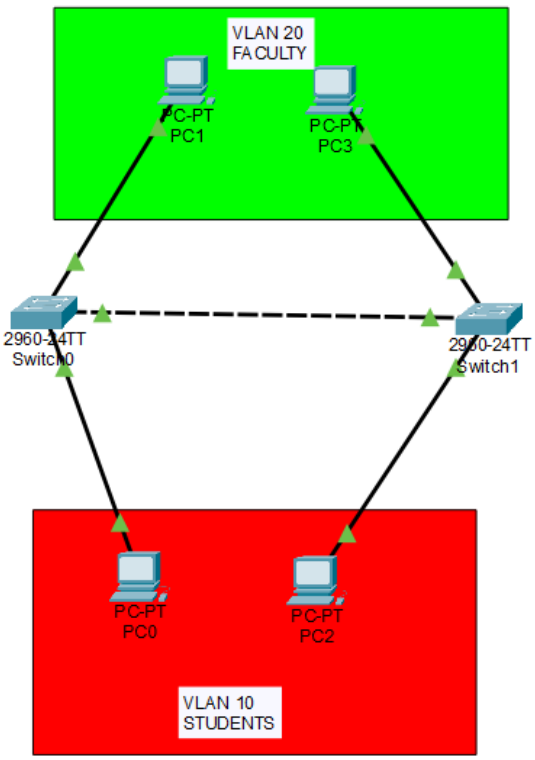
show interfaces trunk

17. Test PC0 → PC2 using ping.
→ Successful, because both are in VLAN 10.

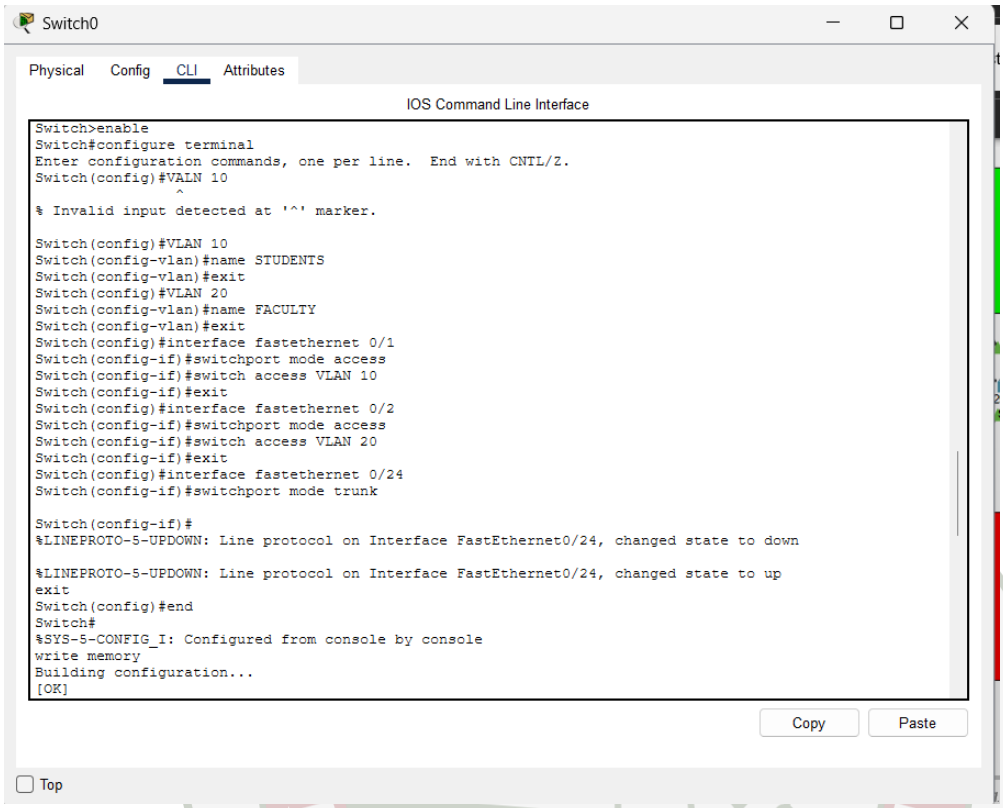
18. Test PC1 → PC3 using ping.
→ Successful, because both are in VLAN 20.

19. Test PC0 → PC1 using ping.
→ Fails, because they belong to different VLANs.

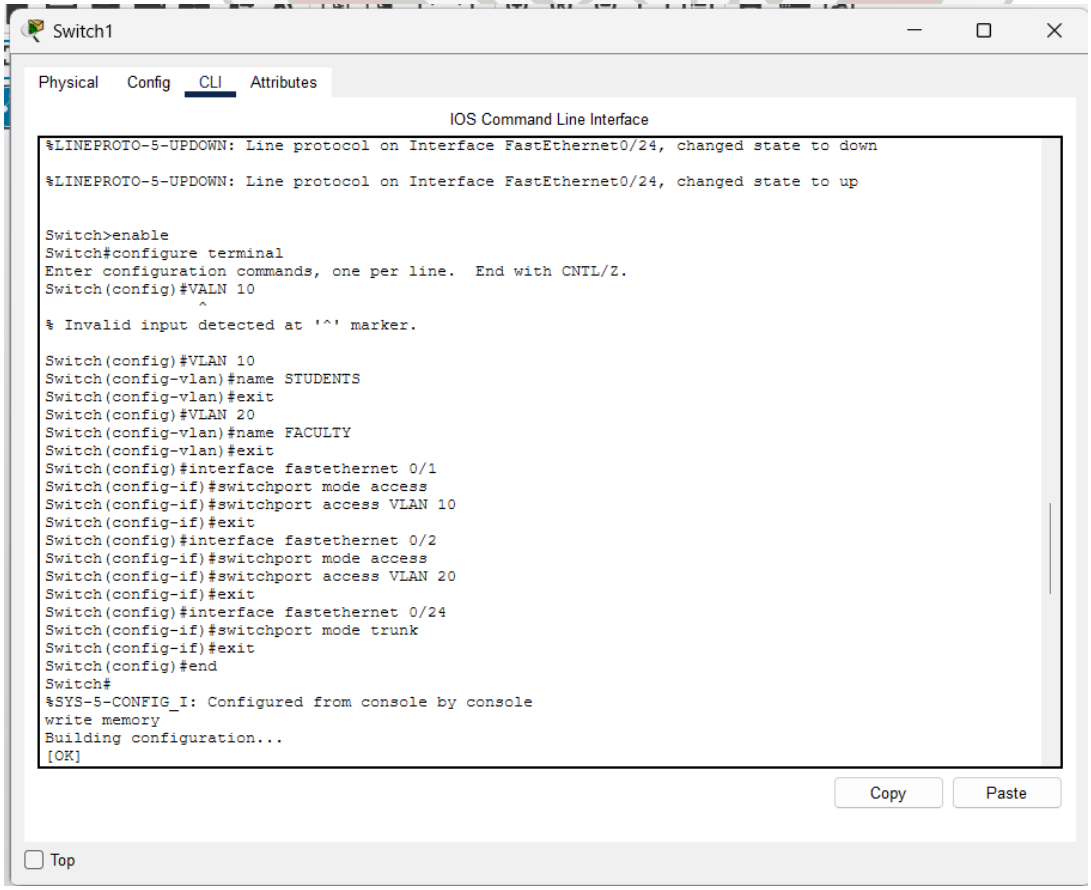
NETWORK TOPOLOGY



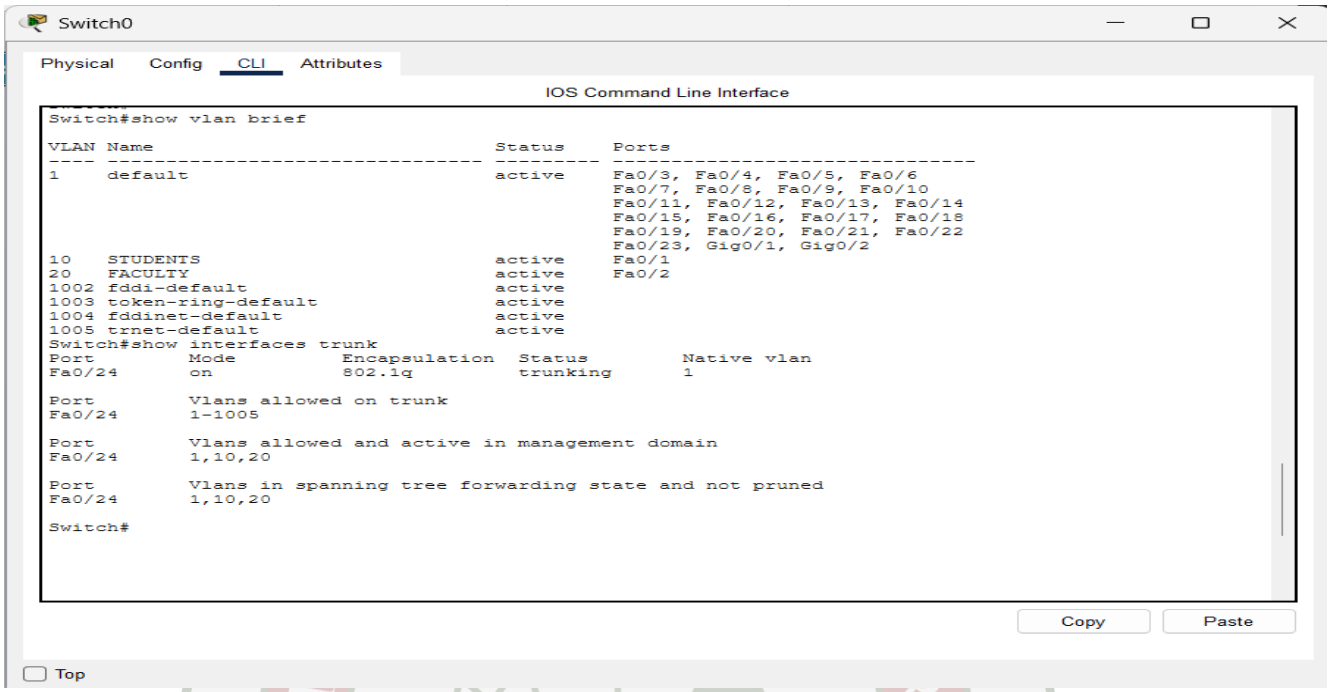
SWITCH 0



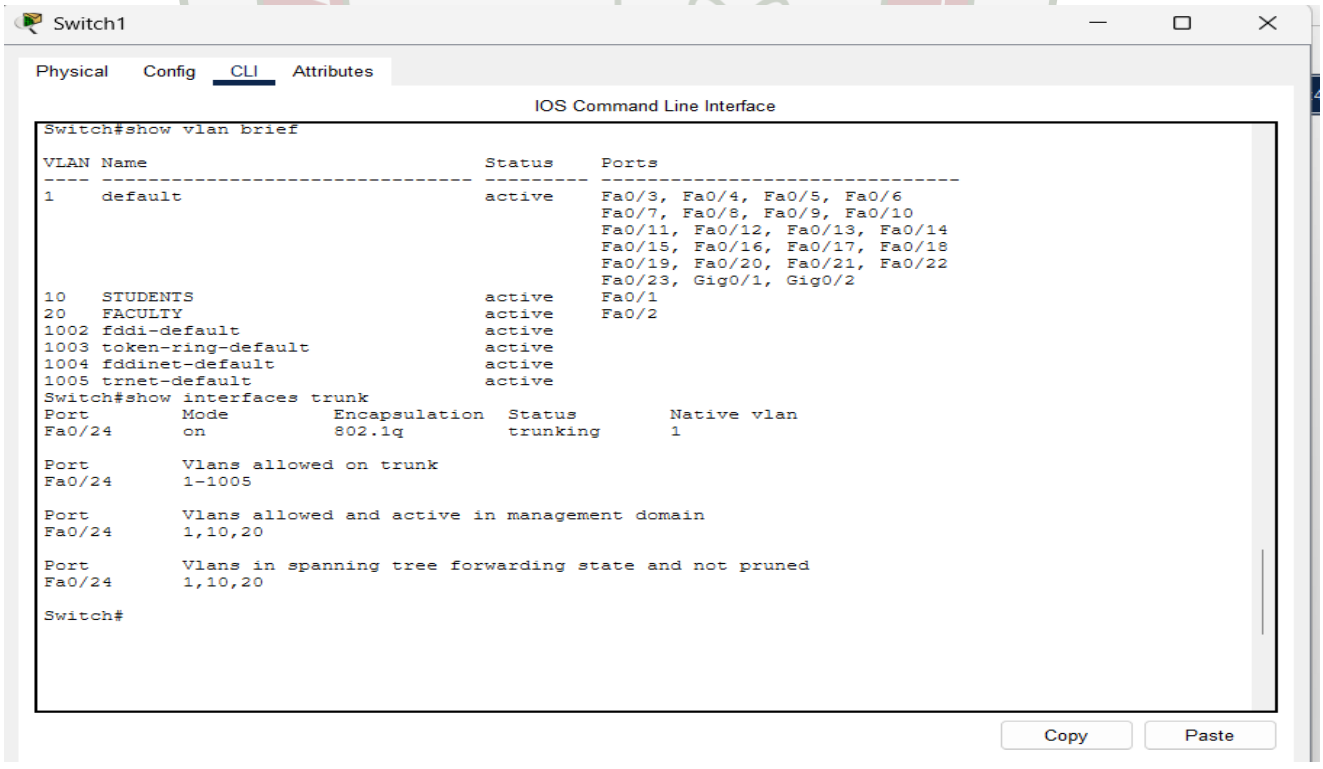
SWITCH 1



SWITCH 0: SHOWS VLAN AND TRUNK



SWITCH 1:SHOWS VLAN AND TRUNK



OUTPUT:

Fire	Last Status	Source	Destination	Type	Color	Time
	Successful	PC1	PC3	ICMP		0.0
	Failed	PC1	PC0	ICMP		0.0
	Successful	PC3	PC1	ICMP		0.0
	Failed	PC3	PC2	ICMP		0.0

Justification:

Using VLANs across two switches allows users belonging to the same logical group to communicate even when they are connected to different physical switches. The trunk link carries traffic from multiple VLANs between the switches.

VLANs provide:

- Network segmentation between different groups.
- Reduced broadcast traffic by creating separate broadcast domains.
- Improved security through logical isolation.
- Better network management and scalability.
- Efficient use of multiple switches without requiring a separate physical network for each group.

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